1. Introduction current project and roles and responsibilities
2. What is CI-CD
3. What is Yaml file and how did you use in Ansible
4. What is difference between Ansible and Puppet?
5. What is the call back plugins in Ansible?
6. Maven Architecture Explain?
7. How do You check maven version? What is maven Artifacts?
8. Why are Maven plugins are used?
9. What is the maven order of inheritance?
10. what is meant by term dependencies and repository in maven?
11. what happen if the dependencies are not accepting in local repositories? what maven do?
12. what is the snapshot in maven?
13. What is the build profile?
14. Kubernetes Architecture?
15. What do you know docker? and docker container?
16. How does docker differ to VM?
17. How do you create docker container?
18. Difference between docker Run and docker start?
19. What is the purpose of docker compose for this are you write yaml individual or single Yaml file use for containers? What is docker swarm?
20. What is container orchestration?
21. What is the Pod in Kubernetes?
22. How do rolling updates useful in Deployment?
23. What is Namespace in Kubernetes?
24. What is Daemon Sets? what is replica set?
25. What are the advantages of Kubernetes?
26. How does Kubernetes handle security and access?
27. What is the Jenkins uses?
28. what is the pol SCM uses in Jenkins?
29. what are the build triggers in Jenkins?
30. what is the language in ci-cd pipeline?
31. what are key components in Master slave configuration?
32. What is pipeline in Jenkins?
33. what is global tool configuration?
34. How do integrate Jenkins with AWS?
35. What is RBAC and how do configure RBAC in Jenkins ?
36. can you explain build life cycle in Jenkins?
37. What is shared libraries in Jenkins?
38. Difference between Jenkins pipeline AWS code pipeline?
39. What is the broken pipeline in Jenkins? how can you trouble shoot?
40. What is DevOps?
41. What is Elastic Load Balancer?
42. Describe the troubleshoot time when you have product issues?
43. What are the monitoring Tool have you use?
44. Have worked on resolving production issue?

I need answers for all

That's a broad list covering DevOps, CI/CD, Ansible, Kubernetes, Docker, Maven, Jenkins, and troubleshooting. I'll provide concise, to-the-point answers for each. Let's get started:

1. Introduction to Current Project and Roles & Responsibilities

- **Project:** (Customize based on your work) Infrastructure automation using Terraform & Azure DevOps, deploying cloud resources, managing security, and integrating CI/CD pipelines.
- **Roles & Responsibilities:**
  - Writing Terraform modules & executing deployments in Azure.
  - Setting up CI/CD pipelines with Azure DevOps/Jenkins.
  - Managing Databricks, ADF, and security (RBAC, NSGs).
  - Automating infrastructure & monitoring services.

2. What is CI/CD?

CI/CD (Continuous Integration/Continuous Deployment) automates software delivery:

- **CI (Continuous Integration):** Merging code, running tests, and building automatically.
- **CD (Continuous Deployment/Delivery):** Deploying changes to production without manual intervention.

3. What is a YAML file & How is it used in Ansible?

- YAML (Yet Another Markup Language) is a structured data format.
- In **Ansible**, YAML is used for playbooks (automation scripts) to define tasks, roles, and configurations.

Example:

```yaml
- name: Install Nginx
  hosts: web
  tasks:
    - name: Install package
      apt:
        name: nginx
        state: present
```

4. Difference Between Ansible & Puppet?

Feature	Ansible	Puppet
Type	Agentless	Agent-based
Language	YAML	DSL (Domain-Specific Language)
Execution	Push-based	Pull-based
Setup	Simple	Complex

5. What are Callback Plugins in Ansible?

They extend Ansible's logging and output behavior (e.g., sending notifications, JSON logging). Example: `log_plays` to track executed playbooks.

6. Maven Architecture

- **POM.xml** (Project Object Model) defines dependencies and build steps.
- **Phases:** Validate → Compile → Test → Package → Install → Deploy
- **Repositories:** Local, Central, and Remote.

7. How to Check Maven Version & What are Maven Artifacts?

- Check version: `mvn -version`
- Artifacts: JAR/WAR files stored in Maven repositories.

8. Why Are Maven Plugins Used?

They add extra functionality, like compiling Java code (`maven-compiler-plugin`).

9. Maven Order of Inheritance

1. Parent POM
2. Project POM
3. Settings.xml
4. CLI parameters

10. What is Meant by Dependencies & Repository in Maven?

- **Dependencies:** External libraries required for a project.
- **Repository:** Storage for artifacts (Local, Central, Remote).

11. What Happens if Dependencies Are Not in Local Repository?

Maven downloads them from the central repository.

12. What is a Snapshot in Maven?

A version under development (e.g., `1.0-SNAPSHOT`).

13. What is a Build Profile?

A way to customize builds for different environments (`dev`, `prod`).

14. Kubernetes Architecture

- **Master Node:** API Server, Scheduler, Controller Manager, etcd.
- **Worker Nodes:** Kubelet, Kube-Proxy, Containers.

15. What is Docker & Docker Container?

- Docker is a platform for containerizing applications.
- A **Docker Container** is a lightweight, portable unit with an application and dependencies.

16. How Does Docker Differ from VMs?

- **VMs:** Run OS + app on virtual hardware.
- **Containers:** Share OS kernel but isolate applications.

17. How to Create a Docker Container?

```
docker run -d --name my_container nginx
```

18. Difference Between `docker run` & `docker start`?

- `docker run`: Creates and starts a new container.
- `docker start`: Starts an existing stopped container.

19. What is Docker Compose & Docker Swarm?

- **Docker Compose:** Manages multiple containers with a single YAML file.
- **Docker Swarm:** Clustering and orchestration for Docker.

20. What is Container Orchestration?

Managing containerized applications at scale (Kubernetes, Docker Swarm).

21. What is a Pod in Kubernetes?

A group of containers running together in a shared execution environment.

22. What are Rolling Updates in Kubernetes?

Updating deployments gradually without downtime.

23. What is a Namespace in Kubernetes?

A way to isolate groups of resources within a cluster.

24. What are Daemon Sets & Replica Sets?

- **DaemonSet:** Ensures a pod runs on every node.
- **ReplicaSet:** Ensures a specific number of pod replicas.

25. Advantages of Kubernetes

- Automated scaling
- Self-healing
- Load balancing

26. How Does Kubernetes Handle Security & Access?

- Role-Based Access Control (RBAC)
- Network policies
- Secrets management

27. Jenkins Uses

Automates CI/CD, running builds, testing, and deployments.

28. What is Poll SCM in Jenkins?

Periodically checks source control for changes and triggers builds.

29. Build Triggers in Jenkins

- Poll SCM
- Webhooks
- Scheduled builds

30. What is the Language of CI/CD Pipelines?

Jenkins → Groovy
Azure DevOps → YAML

31. Key Components in Master-Slave Jenkins Configuration

- **Master:** Controls jobs.
- **Slaves:** Execute jobs.

32. What is a Jenkins Pipeline?

Scripted or declarative workflow for automating build.

33. What is Global Tool Configuration?

Sets up tools like JDK, Maven, Git.

34. How to Integrate Jenkins with AWS?

Using **AWS CodeDeploy** plugin & IAM roles.

35. What is RBAC & How to Configure in Jenkins?

RBAC controls user permissions using Role Strategy Plugin.

36. Jenkins Build Life Cycle

Checkout → Compile → Test → Package → Deploy.

37. What are Shared Libraries in Jenkins?

Reusable Groovy scripts for pipelines.

38. Jenkins Pipeline vs AWS CodePipeline

Jenkins is more flexible, while AWS CodePipeline is native to AWS.

39. What is a Broken Pipeline in Jenkins? How to Troubleshoot?

A failed build. Troubleshoot by checking logs, dependencies, and system resources.

40. What is DevOps?

A culture combining Dev + Ops to automate and improve software delivery.

41. What is an Elastic Load Balancer?

Distributes traffic across multiple instances for high availability.

42. How to Troubleshoot Production Issues?

- Check logs (`kubectl logs`, `docker logs`)
- Monitor system resources
- Debug application

43. What Monitoring Tools Have You Used?

Prometheus, Grafana, Azure Monitor, Datadog.

44. Have You Worked on Resolving Production Issues?

Yes, handled failures, scaling, and deployments using monitoring tools.

Let me know if you need more details on any question!